

Calculus 2 Aug–Nov 2026 Quiz 1 2026-09-03 12:00–13:00

- Total: 60 marks. Contribution towards your grade = (your marks)/4.
1. (a) (8 marks) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Show the graph of f , $\Gamma_f := \{(x, f(x)) \mid x \in [0, 1]\}$ has measure 0.
(b) (7 marks) The unit circle in $\mathbb{R}^2$ has measure 0.
 2. Let $A \subseteq \mathbb{R}^n$ be a closed rectangle. Let $f, g : A \rightarrow \mathbb{R}^n$ be bounded functions. Show the following:
 - (a) (7 marks) There exist $M > 0$ such that for all $x, y \in A$, $|f(x)g(x) - f(y)g(y)| \leq |f(x)||g(x) - g(y)| + |g(y)||f(x) - f(y)| \leq M(|f(x) - f(y)| + |g(x) - g(y)|)$.
 - (b) (7 marks) On every closed rectangle S inside A , $\text{osc}_S(fg) \leq M(\text{osc}_S(f) + \text{osc}_S(g))$.
 - (c) (7 marks) For every partition P of A , $U(fg, P) - L(fg, P) \leq M[(U(f, P) - L(f, P)) + (U(g, P) - L(g, P))]$
 - (d) (4 marks) If f and g are integrable, then so is fg .
 3. (20 marks) Let $A \subseteq \mathbb{R}^n$ be a closed rectangle. Let $f : A \rightarrow \mathbb{R}$ be a bounded function such that for all positive integers n , $\{x \in A : |f(x)| \geq \frac{1}{n}\}$ has content 0. Show that $\int_A f = 0$.